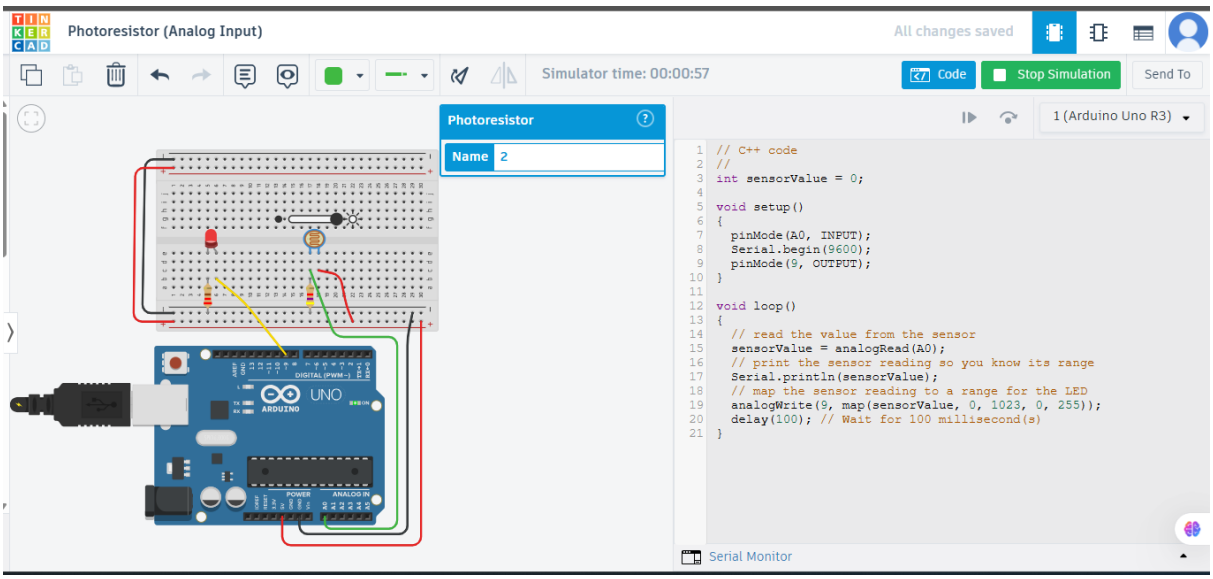
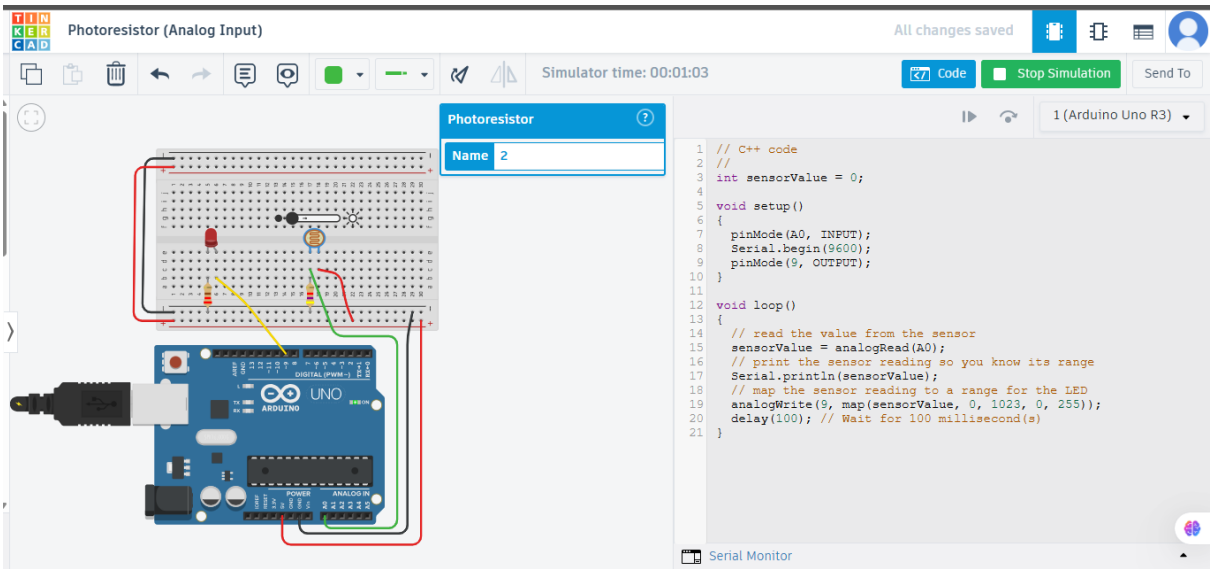
Sensor-Based Simulation

- Simulate an IoT system using Tinkercad/Proteus (no hardware required).
- Example: LED on/off using a sensor (Temperature/Light/Motion).
- Submit simulation screenshots + code explanation.

SIMULATION:





CODE:

```
// C++ code
//
int sensorValue = 0;
void setup()
{
  pinMode(A0, INPUT);
  Serial.begin(9600);
  pinMode(9, OUTPUT);
}
void loop()
{
  // read the value from the sensor
  sensorValue = analogRead(A0);
  // print the sensor reading so you know its range
  Serial.println(sensorValue);
  // map the sensor reading to a range for the LED
  analogWrite(9, map(sensorValue, 0, 1023, 0, 255));
  delay(100); // Wait for 100 millisecond(s)
}
```

EXPLANATION:

1. Variable Declaration

int sensorValue = 0;

- A variable named sensorValue is created
- It stores the **analog value read from the sensor (LDR/potentiometer)**

2. Setup Function

```
void setup()
{
  pinMode(A0, INPUT);
  Serial.begin(9600);
  pinMode(9, OUTPUT);
}
```

pinMode(A0, INPUT);

- Sets **Analog Pin A0** as input
- Sensor (LDR) is connected here

Serial.begin(9600);

- Starts **serial communication**
- Used to display sensor values in **Serial Monitor**

pinMode(9, OUTPUT);

- Sets **Pin 9 as output**
- LED is connected here

3. Loop Function (Main Working Part)

```
void loop()
{
```

- This runs **continuously**

Step 1: Read Sensor Value

```
sensorValue = analogRead(A0);
```

- Reads analog value from sensor
- Range: **0 to 1023**
    - 0 → Dark / Low input
    - 1023 → Bright / High input

Step 2: Print Value

```
Serial.println(sensorValue);
```

- Displays sensor value in **Serial Monitor**
- Helps in debugging and understanding range

Step 3: Control LED Brightness

```
analogWrite(9, map(sensorValue, 0, 1023, 0, 255));
```

This is the **most important line**

map() Function:

- Converts one range into another

```
map(sensorValue, 0, 1023, 0, 255)
```

- Sensor range → **0–1023**
- PWM range → **0–255**

analogWrite(9, value)

- Sends PWM signal to LED
- Controls **brightness**

Step 4: Delay

delay(100);

- Waits for **100 milliseconds**
- Prevents too fast updates